

Compact isolation: think in the gap, not around it

A common mistake is to look for large or exotic solutions—extensive EBG lattices, tall 3D structures, or bulky chokes, when only a few millimetres of real estate are available. In practice, compact isolation structures placed directly in that gap are easier to deploy and integrate.

In our 3.5GHz mutual-coupling work, the parallel coupled-line resonator (PCR) fits entirely between two closely spaced patches on FR-4. The element functions as a local band-stop filter at the operating frequency, so currents attempting to flow between patches encounter high impedance in that narrow region.

Key lessons from deploying and testing these kinds of structures:

- Treat the gap between antennas as a functional area, not a wasted board edge
- Design isolation elements that share the same planar technology and stackup as the antennas
- Aim for resonant or slow-wave behaviour within your band of interest, rather than broad, heavy structures

Case study: PCR-based isolation in a compact 3.5GHz array

One representative design uses a two-element patch array operating at 3.5GHz in the 5G band on a 1.6mm FR-4 substrate. To reflect realistic enclosure and circuitry constraints, the patches are placed only about 6mm apart edge-to-edge, roughly 0.07 wavelengths at 3.5GHz, far tighter than the textbook 0.5λ spacing.

The design was fabricated and tested using a Keysight FieldFox microwave analyser with careful calibration and stable cables.

Without isolation, measurements showed strong coupling between elements and noticeable degradation in efficiency and front-to-back ratio. Introducing a compact PCR

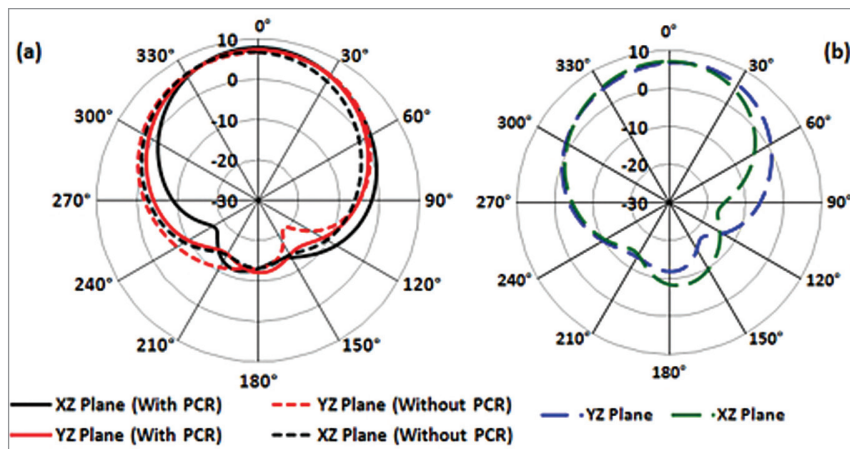


Fig. 2: Radiation patterns and gain: (a) simulated radiation characteristics at 3.5GHz with and without PCR and (b) measured radiation characteristics at 3.5GHz with PCR

in the gap significantly improved performance:

- The PCR consists of pairs of closely spaced microstrip lines, tuned so the structure acts as a band-stop at 3.5GHz
- Two such resonators, cascaded in series, form a high-order stopband within a footprint only a few millimetres wide
- The entire structure remains planar and can be fabricated in the same PCB process as the antennas

Measured S-parameters on the fabricated prototype showed isolation improvements of tens of decibels at the design frequency, with at least double-digit gains maintained across the operating band. Radiation patterns stayed essentially intact, and the array's gain and efficiency improved because less energy leaked into unwanted directions.

The important point is not the exact numbers, but that one narrow isolation element in the gap made a tight-spaced array behave more like a comfortably spaced one.

Consulting lessons: patterns that keep repeating

Across different projects—small-cell CPEs, lab test arrays, and early 5G prototypes—the same patterns keep appearing:

- Most real designs are space-constrained. We rarely have the

luxury of ideal spacings or large isolation walls.

- Planar, compact structures win. Solutions implemented as narrow resonators, slots, or modified grounds are easier to integrate and cheaper to manufacture.

- Isolation must coexist with matching and pattern control. A decoupling element that 'fixes' S21 but destroys S11 or ruins the beam pattern is not usable.

From these experiences, a few practical rules emerge:

1. Always design and characterise a baseline array first, then add isolation elements and compare.

2. If you cannot change overall spacing, work on the field distribution in the gap—via resonators, slots, or carefully placed stubs.

3. Do not over-complicate the structure; complex patterns are sensitive to tolerances and often hard to debug in production.

Design tips for adding compact isolation

For EFY readers wanting to put these ideas into practice, here are some design tips that have worked well in my projects:

- Start simple: Use a 2-element array as a 'lab bench' to prototype your isolation element before extending to larger arrays.
- Size the isolation element to your band: Whether you use